

Assignment-1

Digital VLSI Design

I. SRAM cell design and characterization

To design and characterize the basic 6T SRAM cell as shown in figure 2. The concept and working of SRAM is given here.

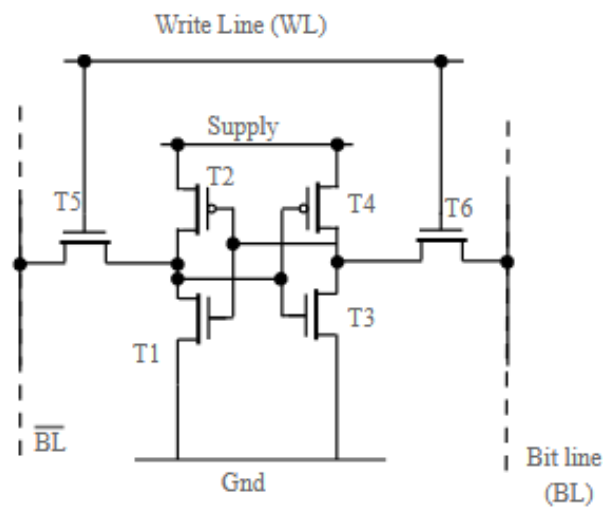


Figure 1: SRAM 6T cell

Recalling that the wordline and bitlines are held at VDD during a read, Figure 2 shows how to extract the read static noise margin (SNM) of the cell. First, the feedback from the cross coupled inverters is broken. Next, the VTC of the “inverter” formed by half of the SRAM cell is found by sweeping V_1 (the inverter’s input) from 0 to VDD and measuring V_2 (the inverter’s output). This plot is then used to construct the “butterfly plot” that is representative of the two halves of the cell driving each other. The read SNM is the side length of the maximum possible square that can fit inside of the butterfly plot. You do not have to calculate the size of this maximum square, but you should submit the butterfly plot (generated using SPICE tool) that graphically indicates the SNM. You should also measure the worst-case voltage rise in the SRAM cell during a read (i.e., the value of V_2 when V_1 is at VDD) and provide that value in your report.

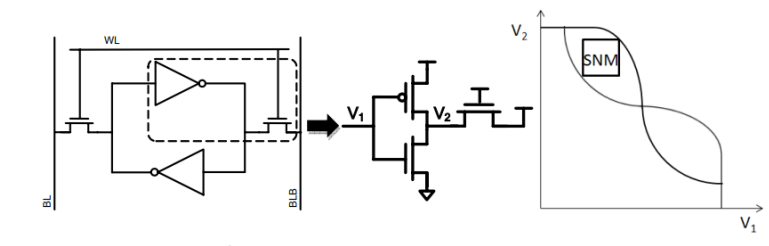


Figure 2: SRAM read static noise margin

During a write, VDD is applied to the wordline, and the value to be written into the memory cell is driven onto the bitlines. Thus, Figure 3 shows how to extract the write noise margin (WNM) of the cell. Again, the feedback from the cross coupled inverters is broken, and the VTC of the “inverters” are measured. Note however that in this case, the VTCs of the two halves of the SRAM are no longer the same (since one of the bitlines is driven to

0V, and the other to VDD). These VTCs are used to create a butterfly plot, and the WNM is the side length of the largest square that can fit inside of the butterfly plot. You do not have to calculate the WNM, but you should generate the butterfly plot (again using SPICE) and graphically indicate the WNM. You should however measure the worst-case cell voltage during a write (which is found from measuring V_1 when V_2 is at 0V).

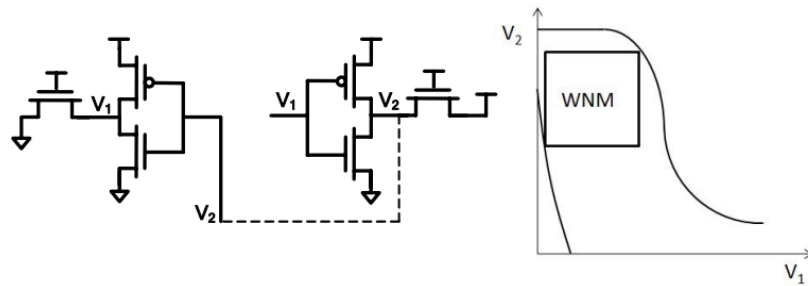


Figure 3: SRAM Write noise margin